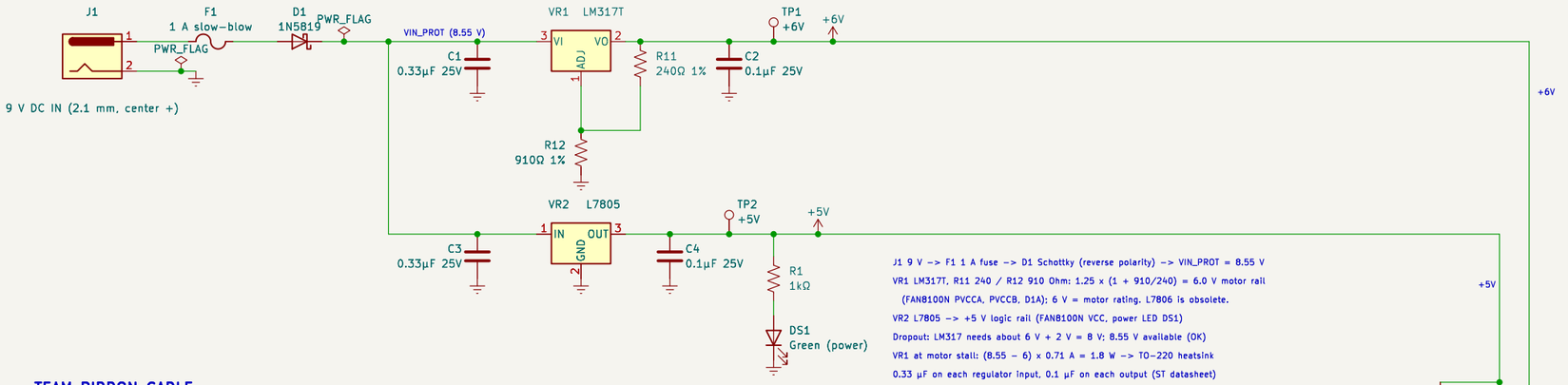
POWER: 9 V IN -> +6 V MOTOR RAIL / +5 V LOGIC RAIL

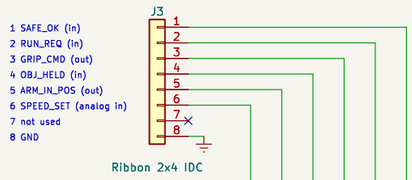


J1 9 V -> F1 1 A fuse -> D1 Schottky (reverse polarity) -> VIN_PROT = 8.55 V
VR1 LM317T, R11 240 / R12 910 Ohm: 1.25 x (1 + 910/240) = 6.0 V motor rail
(FAN8100N PVCCA, PVCCB, D1A); 6 V = motor rating. L7806 is obsolete.
VR2 L7805 -> +5 V logic rail (FAN8100N VCC, power LED DS1)
Dropout: LM317 needs about 6 V + 2 V = 8 V; 8.55 V available (OK)
VR1 at motor stall: (8.55 − 6) x 0.71 A = 1.8 W -> TO-220 heatsink
0.33 µF on each regulator input, 0.1 µF on each output (ST datasheet)
Curiosity Nano is powered from its own USB port (3.3 V on VTG)

DESIGN NOTES

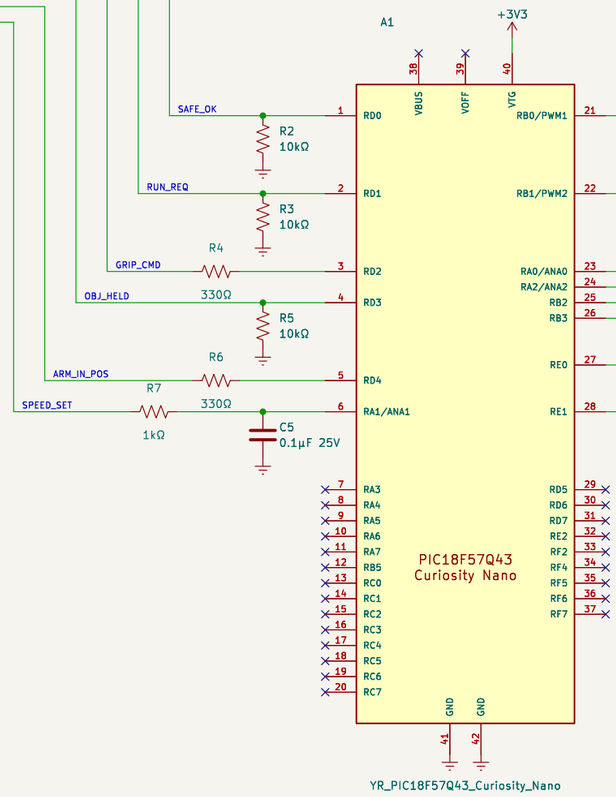
Custom library YR_Custom: symbols YR_PIC18F57Q43_Curiosity_Nano, YR_FAN8100N, YR_Fuse_US;
footprint YR_DIPH-12_W7.62mm_Fin (FAN8100N with GND fin).
Drawn with wires instead of net labels; only power symbols (+6V, +5V, +3V3) and the
earth-ground symbol are global. Text next to each wire names the signal.
US practice (ANSI/IEEE 315, ASME Y14.44): zig-zag resistors, curved-plate polarized
capacitors, S-curve fuse, earth ground; A module, B motor, DS indicator,
VR regulator, F fuse, J connector, SW switch, TP test point.
Resistors 1/4 W 5 %. Capacitor ratings >= 2 x rail voltage.
Rev B: FAN8100N replaces the DRV8874 carrier; position sensor, stop button,
flyback diodes, jumpers, UART and spare-GPIO headers removed; 6 V motor rail.
Rev C: VR1 L7806 (obsolete) -> LM317T set to 6.0 V with R11 / R12.

TEAM RIBBON CABLE



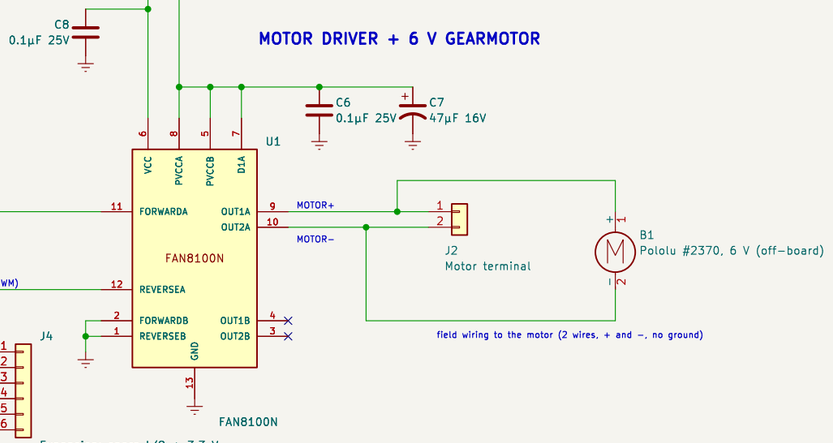
J3 pin order follows the team daisy-chain spec.
10 kΩ pull-downs on SAFE_OK, RUN_REQ, OBJ_HELD:
cable unplugged reads 0 = stop (fail-safe).
330 Ω in series with the two outputs (GRIP_CMD,
ARM_IN_POS) limits current if two boards drive.
1 kΩ + 0.1 µF RC filter (fc = 1.6 kHz) on SPEED_SET.

MICROCONTROLLER (USB powered)



DEBUG: RE0 -> 330 Ω -> LED (4 mA).
RE1 -> button -> GND, internal weak pull-up
(WPUE1 = 1), debounced in firmware.

MOTOR DRIVER + 6 V GEARMOTOR



FAN8100N channel A: FWD=H REV=L forward | FWD=L REV=H reverse | H H brake | L L stand-by
Speed = PWM on RB0 (forward) or RB1 (reverse) while the other input is held low.
100 kΩ pull-downs keep both inputs low (motor off) while the PIC is in reset.
Clamp diodes are built into the FAN8100N, so no external flyback diodes.
Channel B unused: FORWARDB / REVERSEB to GND, OUT1B / OUT2B open.
U1 footprint YR_DIPH-12_W7.62mm_Fin: pads 13 = GND fin; pour copper under the fin.
Bypass: 0.1 µF + 47 µF on PVCC (+6 V), 0.1 µF on VCC (+5 V).

US (ANSI) symbols
9 V input, +6 V motor rail, +5 V logic rail
PIC18F57Q43 Curiosity Nano + FAN8100N H-bridge
Drawn by YR
EGR 304 Embedded Systems — Individual Subsystem

Sheet: /
File: Arm_Motion_Control.kicad_sch

Title: Arm Motion Control — Individual Subsystem Schematic

Size: A3	Date: 2026-10-30	**Rev: C**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1